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Deep learning-based phishing classification framework for accurate detection using optimized URL intelligence

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The proliferation of internet services has exposed customers to phishing attempts that steal sensitive information via false URLs. Intelligent categorization systems are required to tackle dynamic phishing techniques, which defy rule- and signature-based detection. The existing phishing detection systems leverage handmade characteristics or fixed blacklists. That suggests they generalize poorly on zero-day and camouflaged phishing URLs. High false-positive rates and inadequate scalability hamper their performance. It proposes an Adaptive Deep URL Intelligence Network (ADUIN). Deep learning model with optimized URL lexical, host-based, and structural properties. We optimize features using a hybrid relevance-ranking method and train a multi-layer deep neural architecture to understand complicated non-linear phishing patterns. URL intelligence dynamically updates the architecture to resist attack behavior changes. According to experiments on the benchmark phishing dataset, ADUIN is more accurate, exact, and remembers than machine learning classifiers. Zero-day phishing URLs are detected with minimal false alarm rates by the algorithm. The suggested system improves phishing URL classification accuracy, versatility, and intelligence. Real-time online and enterprise security solutions benefit. Under high load, the recommended ADUIN model has 95% classification accuracy, 93% precision, 92% zero-day detection rate, 3.5% false positives, optimal accuracy with 50 features, and 210 ms delay.

Keywords Phishing Detection, Deep Learning, URL Intelligence, Feature Optimization, Cybersecurity, Website Classification

Phishing is the most common and dangerous internet cyberattack. Attackers may mislead tricksters into providing login, financial, and personal information via fake URLs and imitating trusted people[1]. The attack surface has grown significantly, making phishing detection a cybersecurity problem[2]. Quickly evolving internet services, cloud computing, and online communication methods. As phishing methods get increasingly complicated, intelligent and adaptive detection systems that can correctly and scalably identify known and zero-day threats are needed[3].

Evolution of phishing attacks in modern web ecosystems

Phishing assaults are become more sophisticated, targeted, and automated[4]. Blacklists and filters can identify first-generation phishing's fixed URLs and visual signals. Advanced phishing strategies include URL obfuscation, randomized domain creation, short URLs, fast-flux hosting, and HTTPS misuse[5]. Phishing sites are momentarily functioning and undiscovered due to attacks on vulnerable domains and dynamic URLs[6]. Phishing has expanded quicker owing to mobile, social, and cloud computing[7]. Attackers leverage user behavior and contextual information on trusted platforms to scale up malicious URL propagation[8]. PhaaS models have reduced attacker entry costs, allowing large campaigns with little technological skill[9]. This evolving assault method demands real-time detection systems that adapt to URL patterns and threat behavior[10].

PhiUSIIL uses multi-dimensional feature extraction, including lexical analysis, host-based features, and URL structural indicators, to identify harmful URL patterns in big datasets. StealthPhisher employs adaptive machine learning and behavioral URL features to detect subtle phishing signs that circumvent standard detection tools. Integrating these frameworks into the introductory discussion clarifies the evolving phishing detection

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research landscape, shows the shift from feature-based models to adaptive intelligence-driven approaches, and strengthens the contextual foundation for the proposed ADUIN[11].

Limitations of conventional URL-based detection techniques

Tradition URL-based phishing detection uses handwork, blacklists, and signature rules. These methods work against known phishing patterns but not zero-day attacks[12]. Blacklist instances must be updated periodically and cannot disclose new phishing domains[13]. Manually produced characteristics seldom capture the multidimensional and non-linear structure of modern phishing URLs[14]. Rule-based classifiers rely largely on them. In real life, traditional machine learning models have limited scalability and high false-positive rates. Fixed feature distributions prevent these models from adapting to threat distributions. Thus, they mislabel safe and rare URLs as dangerous, decreasing user trust and system usefulness[15]. Due to its static effectiveness, new dynamic and intelligent detection methods that can react to changing phishing trends are required.

Motivation for adaptive deep URL intelligence

Due to weak phishing detection methods, deep learning-based solutions that automatically extract hierarchical and non-linear feature representation from URL data are suggested. Deep neural networks imitate fine-grained correlations between lexical, host-based, and structural URL features better than standard methods. With optimum URL knowledge, deep learning models may increase generalization and resistance to obfuscated and zero-day phishing.

Moreover, adaptive learning mechanism is necessary to be applicable to dynamic threat environments to be effective in the long run. It is possible to incorporate continuous updates to URL intelligence to make the detection framework adapt to attacker strategies and minimize false positives and ensure a high detection rate. The result of this motivation is the proposed Adaptive Deep URL Intelligence Network (ADUIN) that integrates the optimization of feature selection with the deep learning to provide a scalable, true, and robust phishing detector that can be used in real-time web-based and enterprise-wide security systems.

Contributions of this paper

- An innovative Adaptive Deep URL Intelligence Network (ADUIN) that is able to learn intricate non-linear phishing behaviors.
- Optimized URL feature intelligence mechanism that incorporates lexical, host-based and structural features.
- Extensive experimental results showing better accuracy, precision, and recall than current methods.

This article continues as follows. Section 2 discusses phishing detection and intelligent URL analysis background and research. Section 3 describes the phishing URL categorization issue formulation and threat model. The deep learning–based phishing detection framework and system workflow are shown in Sect. 4. Section 5 illustrates how optimal feature selection and relevance-ranking discover the most discriminative URL intelligence properties. Section 6 describes the Adaptive Deep URL Intelligence Network (ADUIN) architecture, including feature representation and attention-based learning. Section 7 describes the approach, dataset configuration, and validation plan for the experiment. In Sect. 8, results and performance analysis are compared to baseline models. Section 9 concludes and suggests future study on intelligent phishing detection systems.

Background and related work

The proposition of a hybrid phishing URL detection system is suggested as the Logistic-SVC-Decision Tree Voting Model (LSD-VM) on a URL-based dataset of more than 11.000 phishing and legitimate URLs. After preprocessing there are various machine learning classifiers that are tested. The LSD-VM uses the Logistic Regression, Support Vector Classifier, and Decision Tree by soft and hard voting. The cross-validation and the optimization of the Canopy technique are used to select the features leading to the high accuracy, the precision, the recall and the F1-score, as well as the specificity[16].

The proposed is a Cyber Threat Intelligence Two-Stage Ensemble Model (CTI-TSEM) to detect malicious URLs. The use of CTI based features derived on a Google search and the WHOIS database improves classification. Random Forest is used to do preclassification, and final decision-making is sent to a Multilayer Perceptron on the probabilistic results of the first stage. CTI-TSEM enables far better detection accuracy and minimizes false positives than traditional URL-based methods of detection through the replacement of majority voting by neural aggregation[17].

One phishing detection method is introduced dubbed Hyperlink-Based Deep Neural Network (HB-DNN) wherein 30 hyperlink-associated features of an HTML source code are used, as opposed to the content of web pages. These characteristics are also utilized to train a supervised Deep Neural Network that is optimized using the Adam optimizer and Listwise ranking strategy. HB-DNN is accurate and robust and works well with large size datasets, it beats the traditional classifiers like SVM, AdaBoost and AdaRank[18].

The XAIAOA-WPC (Explainable AI with Aquila Optimization Algorithm Web Phishing Classification) is an explainable phishing detection framework that is created through the combination of preprocessing, Harris Hawks Optimization-based feature selection, and Multi-Head Attention LSTM classifier. Aquila Optimization is better at detecting, and LIME is interpretable. XAIAOA-WPC is designed to support Cyber-Physical systems and achieved 99.29% accuracy on the benchmark data, thus it has good classification results with explainable decision-making[19].

The paper proposes a phishing URL detection method (DM-PSO-PUD) (Data Mining with Particle Swarm Optimization for Phishing URL Detection) that works with structural URL features generated by data mining rules. The feature selection and classification are done using Particle Swarm Optimization. This optimization

based framework maximizes true positive rates and minimizes processing time and is therefore appropriate in real time and large scale phishing URL detection[20].

Optimization and ensemble learning–based phishing detection models

The presentation of a Fine-Tuned Machine Learning Framework (FT-MLPD) is introduced to optimize the detection performance by balancing the data, optimizing the hyperparameters, and selecting the features. There are eight machine learning classifiers that are tested on two popular phishing datasets. The findings indicate that hyperparameter optimization and feature selection lead to high accuracy, and the accuracy rates of Random Forest and Gradient Boosting are higher than 98%[21].

Phishing URL Detection using Similarity Index and Incremental Learning (Phishing URL Detection) is a phishing detection framework that is created to resist visually deceitful attacks like homograph and punycode phishing. Incremental learning and similarity indexing allow the constant updating of the models. Testing on more than 235,000 URLs shows high adaptability with accuracy of 99.79% with pre-training and 99.24% with incremental learning[22].

It is proposed that a PSO-SVM Feature Selection Model of URL Defacement Detection (PSO-FS-UDD) can be used with the help of Particle Swarm Optimization to choose the best features based on the ISCX-URL-2016 data set. A variety of classifiers are tested among them, SVM, Decision Tree, Random Forest, and Naive Bayes. Findings have shown PSO-FS-UDD + Random Forest to have better accuracy, precision, and F-measure in identifying attacks of URL defacement attacks[23].

To optimally phish a better classification accuracy, a PSO-Based Feature Optimization Phishing Detection Framework (PSO-FOPDF) is created to optimize the selection of features. Particle Swarm Optimization is used on machine learning and the neural network models with two phishing benchmark datasets. The results of the experiments indicate significant increases in accuracy, sensitivity, specificity and Matthews correlation coefficient with the accuracy as high as 97.81[24].

The HawkPhish-DNN phishing detection framework combines the Harris Hawk Optimization and Deep Neural Networks. The model derives the URL length and entropy features, eliminates redundant qualities in the model, and uses Sigmoid and ReLU layers to perform streamlined classification. Multi-objective HHO minimizes false alarms but maximizes accuracy. HawkPhish-DNN attains a false positive rate of 0.2 and 99.6% accuracy[25].

Deep learning and explainable AI-driven advanced detection frameworks

The hybrid phishing URL detection system named IWQPSO-FMRCNN (Improved Weighted Quantum Particle Swarm Optimization with Fast Mask RCNN) has been proposed without the need of webpage content or third-party services. IWQPSO is used to optimize FMRCNN parameters to enhance the detection performance. The results of the experimental evaluation are high as they reached 99.3% accuracy and high recall, which proves the effectiveness of the integrated evolutionary optimization and deep learning[26].

The use of few 30 URL-based features such as entropy, suspicious terms, brand matching using TF-IDF and structural URL features, is introduced as a Lightweight URL-Based Phishing Detection Framework (LU-PDF). Six datasets are tested on eight machine learning classifiers. The best performance belongs to the Random Forest: the accuracy can be 96.85 without visiting webpages and relying on a third party[27].

A phishing detection tool named as Residual Pipeline Phishing URL Detector (RP-PUD) is suggested based on the lexical features of URL, age of domain and sentiment based features. The extracted features are processed by a residual pipeline that consists of convolutional and inverted residual blocks and then classified by an MLP. Assessment on Kaggle datasets yields an accuracy of 98.29 and has been found to be very resilient to the changing phishing methods[28].

A phishing detection model, referred to as Nature-Inspired SVM Optimization Model (NI-SVM-OM), is designed based on URL-based characteristics and an SVM classifier minimized by Bat, Firefly, Grey Wolf Optimizer and Whale Optimization algorithms. According to experimental results, the Gray Wolf Optimizer can dramatically improve the performance of SVM, and it is better than grid-search optimized Random Forest models[29].

A Genetic Algorithms-based ensemble optimization framework called GA-Optimized Ensemble Phishing Detector (GA-OEPD) is suggested to use Genetic Algorithms to optimize the ensemble classifiers, which include Random Forest, AdaBoost, XGBoost, Bagging, Gradient Boosting and LightGBM. The final detection model is made by ranking and combining optimized classifiers. GA-OEPD has an accuracy of 97.16% that indicates usefulness in the IoT and network security environments[30].

Ehsan Nowroozi et al.[31] suggested the Random Deep Feature Selection Approach to Mitigate Transferable Adversarial Attacks. This paper suggests random deep feature selection to reduce transferability and improve model robustness to adversarial manipulations. This approach protects deep models from poisoning (e.g., label flipping) and exploratory (e.g., DeepFool, BIM, FGSM, I-FGSM, L-BFGS, C&W, JSMA, and PGD) assaults during training and testing, as well as Transfer Learning-Based Adversarial assaults. Considerations include perfect and semi-knowledgeable attackers. This method is tested on the UNSW-NB15 dataset, which includes real-world and synthetic data on a wide spectrum of recent attack behaviors and benign activities. This method improves the target network's effectiveness by over 80% against label-flipping poisoning attacks and over 60% against all major exploratory attacks.

Rahim TAHERI et al.[32] proposed the Feature Selection In Intrusion Detection System. Numerous intrusion detection systems with different assault detection algorithms aim to improve efficiency and accuracy. Developing techniques to improve their efficiency is required. This paper seeks to reduce feature count. All features were analyzed using Cuttlefish algorithm and numbers 3, 5, 10, and 13. We used artificial neural network to evaluate

this research and compared its results to others. The thirteenth characteristic was the most effective and detected nearly all attacks.

Table 1 shows the comparative analysis of phishing detection methods.

Research gaps in zero-day phishing identification

Regardless of the fact that a lot of research has been done on phishing detection, a number of gaps are present. The available methods are mostly based on pre-classified or benchmark datasets, which restrict the real world generalization to new and changing phishing attacks. There is also a tendency to be complicated, time-intensive, or reliant on third-party services when extracting features, making it less applicable in real time. A lot of high-accuracy models are not explainable thus, cannot be applied in security-sensitive environments. Also, little attention is paid to incremental learning, adversarial robustness, and deployment efficiency, indicating that lightweight, adaptive, and explainable phishing detection frameworks are required.

Problem definition and threat model

The deceptive URLs that are used to take advantage of users in phishing attacks are mainly supposed to be authentic web resources but they are fake. Emails, social networking, instant messaging, and hacked websites are phishing URL attack vectors. To stay undetected, attackers exploit high domain creation speed, URL shortening, HTTPS abuse, fast-flux hosting, and rapid infrastructure changes. The attackers encode, abuse subdomains, utilize random tokens, and redirection pathways to block URL features, making static and hand-written feature-based site detection systems worthless. These evasion methods reduce blacklist and rule-driven systems' efficacy, especially zero-day phishing URLs. Thus, identifying malicious URLs in a dynamic and hostile environment is difficult. The best detection system is dynamic, scalable, and able to react to changing attack patterns and feature obfuscation. These objectives require intelligent algorithms that automatically extract and optimize discriminative URL intelligence.

Proposed deep learning-based phishing detection framework

An Adaptive Deep URL Intelligence Network (ADUIN) is proposed to improve phishing detection systems. Optimized URL intelligence and deep learning provide reliable and scalable phishing categorization in this system. Modular workflow includes URL capture, preprocessing, feature extraction, optimization, and adaptive deep learning-based classification. Modules may handle phishing and hidden URLs. Its flexibility, robustness, and low false-positive limit its use in real-time enterprise security, internet gateway, and email filtering.

System overview of adaptive deep URL intelligence network (ADUIN)

Deep learning phishing detection model ADUIN uses multi-dimensional URL intelligence to detect fakes. Deep neural classification network, hybrid feature optimization module, and feature extraction layer compose the architecture. Unlike older systems, ADUIN updates its learning representations with new URL patterns to respond to new threats. It automatically inputs and outputs without static rules or blacklists. This architecture can withstand zero-day phishing attacks and adapt to changing internet environments.

Figure 1 shows the whole Adaptive Deep URL Intelligence Network (ADUIN) phishing URL detection method. Raw URLs are normalized using real-time online traffic, email gateways, or corporate security logs. The peding and normalization is used to decode the URL encodings, eliminate noise, encode components, and standardize the inputs. A streamlined feature extraction engine extracts lexical, host-based as well as structural URL intelligence features. These are further eliminated with the aid of a hybrid relevance-ranking mechanism to combine reduced dimensions and redundancy. The optimized set of features is then passed to the ADUIN deep

No.	Domain	Existing method	Real-time/scalable	Explainable	Efficiency
1	Phishing URL Detection	LSD Voting Model (LR + SVC+DT)	✓	✗	High
2	Malicious URL Detection	CTI Two-Stage Ensemble (RF + MLP)	✗	✗	High
3	Phishing Website Detection	Hyperlink-Based DNN (HB-DNN)	✗	✗	Medium
4	CPS Web Security	XAIAOA-WPC (MHA-LSTM + XAI)	✗	✓	Very High
5	Phishing URL Detection	DM-PSO-PUD	✓	✗	Medium
6	Phishing Classification	FT-MLPD (Tuned ML Models)	✓	✗	High
7	Phishing URL Detection	PhiUSIIL (Similarity + Incremental)	✓	✗	Very High
8	URL Defacement Detection	PSO-FS-UDD	✓	✗	High
9	Phishing Website Detection	PSO-FOPDF	✗	✗	High
10	Phishing Detection	HawkPhish-DNN	✓	✗	Very High
11	Phishing URL Prediction	IWQPSO-FMRCNN	✗	✗	High
12	Lightweight Phishing Detection	LU-PDF (URL-Based RF)	✓	✗	High
13	Phishing URL Detection	RP-PUD (Residual Pipeline + MLP)	✓	✗	High
14	Phishing Classification	NI-SVM-OM (Nature-Inspired SVM)	✗	✗	Medium
15	IoT Phishing Detection	GA-OEPD (GA-Optimized Ensemble)	✓	✗	High

Table 1. Comparative analysis of phishing detection methods.

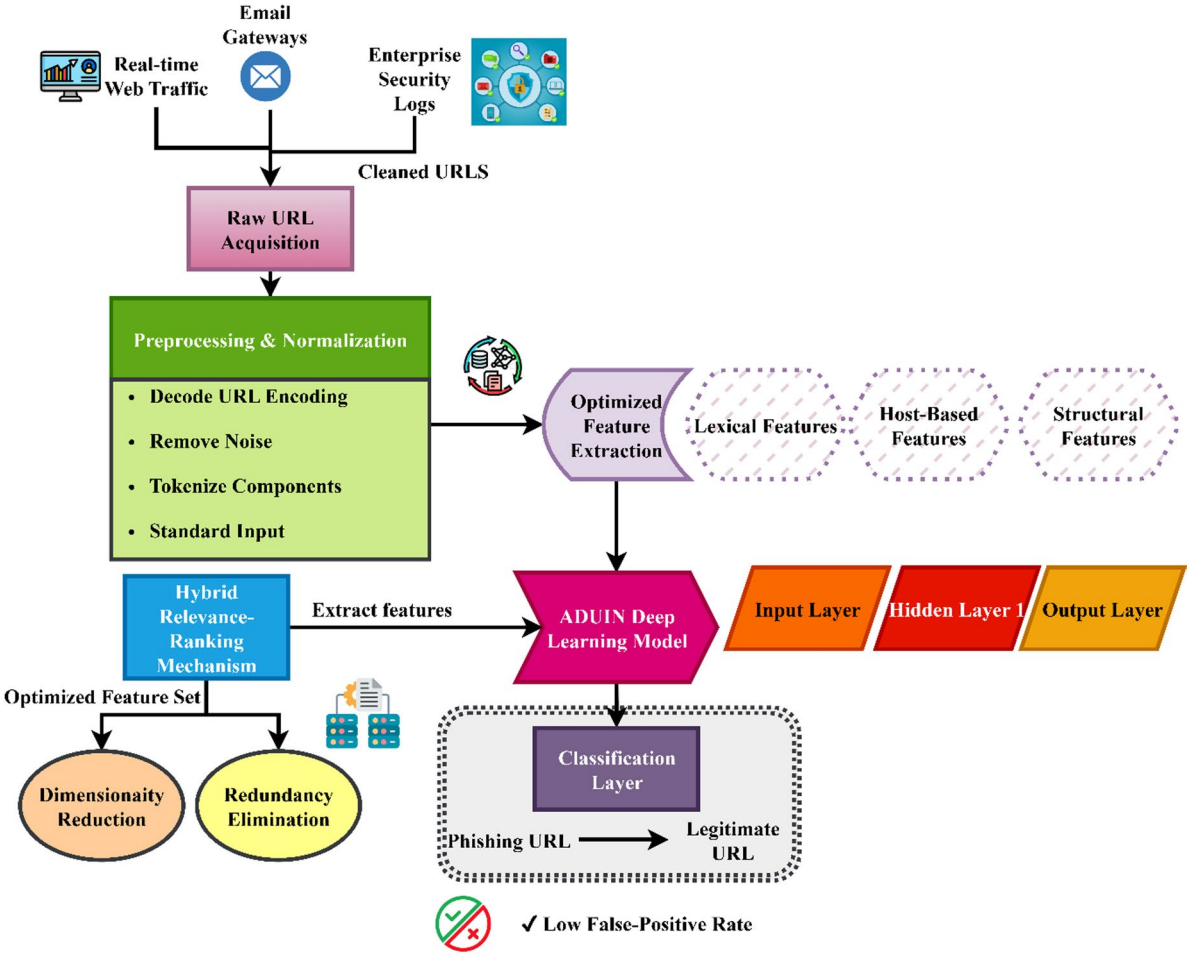


Fig. 1. Overall ADUIN phishing detection framework.

learning model which learns the non-linear phishing patterns. The output of the classification layer is the final decision which is a correct identification of phishing or legitimate URLs with low false-positive rates.

Hybrid feature concatenation fall is expressed in Eq. 1.

$$\text{fall} = \text{fl}\ \text{fh}\ \text{fs} \tag{1}$$

This equation combines multiple feature types. Hybrid representation improves discrimination.

It captures complementary phishing evidence.

fall is the combined feature vector, fl is the lexical feature vector, fh is the host-based feature vector, and fs is the structural feature vector.

Deep-layer linear transformation Z1 is expressed in Eq. 2.

$$\text{Z1} = \text{W1}\ \text{fopt} + \text{b1} \tag{2}$$

This equation performs linear projection. Features are mapped into the latent space. It prepares data for non-linear learning.

Z1 is the linear activation vector, W1 is the weight matrix, fopt is the optimized feature vector, and b1 is the bias vector.

Deep feature refinement h2 is expressed in Eq. 3.

$$\text{h2} = \sigma\text{W2}\ \text{h1} + \text{b2} \tag{3}$$

This equation refines deep representations. Higher-order phishing behaviors are captured.

Depth enhances model expressiveness.

h2 is the refined hidden vector, W2 is the weight matrix, h1 is the previous hidden vector, and b2 is the bias vector.

Output logit computation o is expressed in Eq. 4.

$$\text{o} = (1 - \text{Wo} * \text{h2} + \text{bo}) \tag{4}$$

This equation computes class logits, logits represent raw class confidence. They precede probability estimation.

o is the output logit vector, Wo is the output weight matrix, h2 is the refined hidden vector, and bo is the output bias.

URL data acquisition and preprocessing strategy

The data on URLs are gathered across varied sources which include phishing repositories, legal web traffic and security logs so that the diversity of the data is achieved. Preprocessing is done to normalize the URLs by decoding characters in the encodings, noise removal, and missing attributes and tokenization of the URL components. Maintaining syntactic and semantic content and removing inconsistencies is a priority. This preprocessing step standardizes feature extraction input. Raw URL standardization enhances bias, deep neural network learning, and feature reliability.

In phishing URLs, percent encoding (URL encoding), HTML entity encoding, and hexadecimal representations are utilized to hide harmful tokens. To preserve URL structure, encoded characters like "%2F", "%3A", and "%40" are transformed to their symbols ("/", ":", "@"). For consistent lexical analysis, Unicode escape sequences and mixed encoding patterns are normalized. URL noise elimination targets superfluous delimiters, repetitive special characters, tracking parameters, session IDs, and overly large query strings that do not contribute to phishing semantics. Remove duplicate URLs, reduce whitespace artifacts, convert characters to lowercase, and equalize delimiter use across domain and path segments during preprocessing.

Multi-dimensional URL intelligence feature extraction

URL intelligence is obtained via lexical, host-based, and structural URL analysis. This multilayer method detects surface and contextual phishing. Multiple factors offer the system a global perspective of URL characteristics. Phishing test generality and obfuscation resistance improve with this method. Deep learning algorithms can differentiate lawful and malicious URLs due to their rich numerical features.

The phishing detection work of Algorithm 1 entails extraction of complete features of raw URL. All URLs' lexical (length, entropy, special characters), host-specific (domain age, DNS reputation, IP characteristics), and structural (path depth, subdomains, redirections) features are calculated. Adding a normalized vector of characteristics creates input matrix X. This statistic will provide the deep learning machine rich, discriminative URL representations for phishing trends.

During preprocessing, each URL domain is resolved to its corresponding IP address, after which reputation indicators are obtained from aggregated security intelligence feeds that track known phishing hosts, malicious domains, and suspicious network activity. The reputation score is represented as a normalized numerical feature ranging from 0 to 1, where values closer to 1 indicate higher malicious association based on prior phishing reports, blacklist presence, and abnormal hosting behavior, while values closer to 0 correspond to trusted or previously unflagged IP addresses. Additional indicators such as frequency of malicious domain hosting, blacklist occurrences, and ASN-level risk association are incorporated into the final score through weighted aggregation.

Lexical feature representation

Lexical features characterize syntactic attributes of URLs and they are good in detecting suspicious patterns. These characteristics are the length of the URL, the distribution of characters, the entropy, the amount of special characters, the suspicious keywords and the frequency of the tokens. Obfuscation and randomization techniques tend to have abnormal lexical features in phishing URLs. The system forecasts the signs of an ill intent by numerically modeling these properties. Lexical feature is computationally inexpensive and is important in identifying new phishing URLs that do not have historical and host related information.

The keyword feature is derived from a curated dictionary of commonly abused phishing-related terms observed in credential harvesting URLs and deceptive login pages. The dictionary includes tokens frequently associated with impersonation or account verification prompts, such as login, signin, verify, secure, account, update, banking, confirm, password, payment, webscr, support, security, and authentication. During feature extraction, each URL string is first normalized through lowercase conversion and delimiter tokenization using characters such as "/", "?", "=", "-", and "_". A keyword matching procedure is then applied to identify the occurrence of any dictionary term within URL tokens. The resulting feature representation encodes (i) the presence or absence of suspicious keywords, (ii) the total keyword frequency within the URL, and (iii) the positional occurrence of keywords in domain or path segments.

Host-based and domain intelligence features

Host-based features offer contextual data of the domain properties and the URL hosting environment. These are domain age, WHOIS registration information, IP address reputation, DNS records and hosting stability. Phishing domains can be newer, temporary or can be located on questionable infrastructures. The use of domain intelligence helps the detection system identify between the legitimate long-standing domains and malicious domains. These characteristics make the detection more accurate especially when lexical features do not prove to be enough to detect advanced cases of phishing.

Host-based attributes such as WHOIS metadata, DNS resolution information, and IP reputation indicators are incorporated through a hybrid feature-acquisition strategy that combines pre-computed intelligence caching and lightweight runtime feature extraction. During dataset preparation using the PhiUSIIL Phishing URL Websites Dataset, host-related information is collected offline and transformed into structured numerical representations (e.g., domain age intervals, DNS record existence flags, ASN associations, and IP reputation scores). These attributes are stored in a locally indexed feature repository, allowing constant-time retrieval during inference. At runtime, the ADUIN primarily processes lexical and structural URL features, which are directly extracted

Input: Raw URL dataset U={u1,u2,...,uN}

Output: Feature matrix X∈RN×F

1. Initialize feature matrix X←0N×F
2. for i = 1 to Ndo
3. ui← decode URL, remove noise, tokenize
4. // Lexical features
5. Compute URL length Li, character entropy Ei, special character count Si
6. // Host-based features
7. Retrieve domain age Di, DNS reputation Ri, IP address properties Ii
8. // Structural features
9. Compute path depth Pi, subdomain count SDi, redirection chains RCi
10. Concatenate features: xi=[Li,Ei,Si,Di,Ri,Ii,Pi,SDi,RCi]
11. Insert xi into X[i,:]
12. end for
13. Normalize Xusing min-max or z-score
14. return X

Algorithm 1. Multi-dimensional URL feature extraction.

from the URL string without external queries, while cached host-based indicators are retrieved through in-memory lookup tables or key–value indexing. This architecture eliminates repeated external WHOIS or DNS queries during inference, thereby significantly reducing network overhead. Consequently, the classification pipeline operates as a single-pass feature extraction and neural inference process, enabling an average detection latency of approximately 210 ms under high-load conditions while maintaining the benefit of incorporating host intelligence signals during the feature engineering phase.

Structural and behavioral URL patterns

The structural and behavioral features are used to examine the navigational and hierarchical characteristics of URLs. This comprises of path depth, number of subdomains, redirection paths, use of HTTPS, and irregular URL format. Phishing URLs use too many subdomains, fake directory paths and redirects to cover their motives. Identification of behavioral anomalies that are indicative of phishing is made by capturing these patterns. Lexical and host-based features go hand in hand with structural features, which allow the detection model to distinguish sophisticated evasion techniques employed by advanced phishing campaigns shown in Fig. 2.

The internal architecture of the Adaptive Deep URL Intelligence Network to be applied in phishing classification is shown in Fig. 3. The input layer is then fed with the optimized feature vector and converts numerical URL intelligence features into the neural network. Several deep hidden layers that use non-linear activation functions like ReLU represent the interactions among complex features and latent features of phishing. The network acquires hierarchical representations that facilitate resistant detection of the phishing URLs of obfuscation and zero-day. An adaptive learning module keeps updating network weights with the new URL intelligence observed and and this keeps them immune to changes in attack schemes. Sigmoid or softmax activation are used in the output layer to come up with the final classification decision with high precision and recall.

Second hidden layer transformation z2 is expressed in Eq. 5.

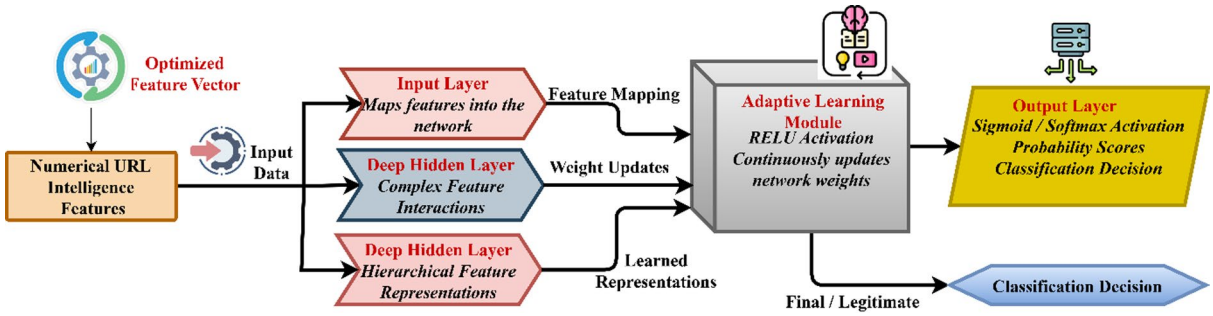


Fig. 2. Adaptive deep URL intelligence network (ADUIN) architecture.

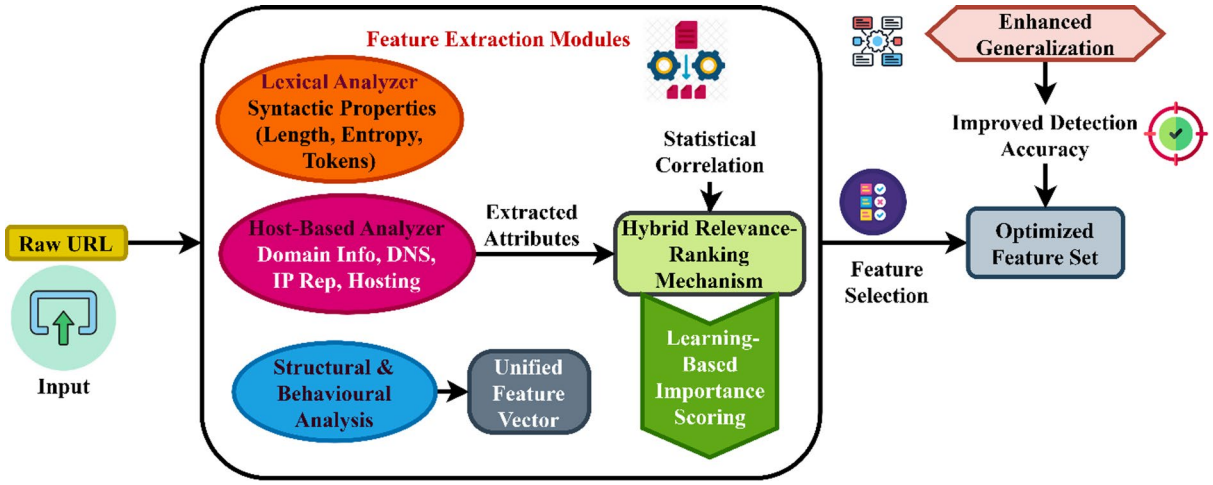


Fig. 3. Optimized URL feature intelligence flow.

$$z2 = w2\ h1 + b2 \tag{5}$$

This equation deepens feature abstraction. Hidden features are recombined hierarchically.

It captures complex phishing patterns.

z2 is the second-layer activation, w2 is the weight matrix, h1 is the previous hidden vector, and b2 is the bias vector.

Deep feature refinement layer z3 is expressed in Eq. 6.

$$z3 = w3\ h2 + b3 \tag{6}$$

This equation refines latent representations. Higher-order phishing interactions are modeled.

Depth improves zero-day detection.

z3 is the third-layer activation, w3 is the weight matrix, h2 is the second hidden vector, and b3 is the bias vector.

Hierarchical feature aggregation hagg is expressed in Eq. 7.

$$hagg = h1 + h2 + h3 \tag{7}$$

This equation aggregates hierarchical knowledge. Multi-level features reinforce decisions.

It improves detection stability.

hagg is the aggregated hidden vector, h1 is the first hidden vector, h2is the second hidden vector, and h3 is the third hidden vector.

Weight update rule Wt + 1 is expressed in Eq. 8.

$$Wt + 1 = \ Wt - \ nt\ \nabla L \tag{8}$$

This equation updates the network weights, and the gradients guide the learning direction. Adaptive updates improve generalization. Wt + 1 is the updated weight matrix, Wt is the current weight matrix, nt is the learning rate, and ∇L is the loss gradient.

Network parameters are iteratively updated according to Wt + 1 = Wt-ηt∇L, where Wtrepresents the model weights at iteration t, ηtdenotes the learning rate, and ∇Lis the gradient of the loss function with respect to the parameters. During model training, URL intelligence features extracted from lexical, structural, and host-based attributes are processed through forward propagation, and the resulting classification error is used to compute gradients via backpropagation. Parameter updates occur at each mini-batch training iteration, with a batch size of 128 URLs per update step, allowing the model to progressively refine feature representations throughout the training process. This iterative optimization continues across 50 training epochs, enabling the network to gradually adjust its internal representations and capture discriminative phishing patterns across diverse URL structures.

Optimized feature selection and relevance ranking

Optimising high-dimensional URL properties enhances model efficiency and accuracy. The framework optimises feature selection to reduce duplicate and redundant characteristics. This lowers deep learning model overfitting and processing costs. The system stabilizes learning and generalization by preserving just the most discriminative characteristics. The framework scales in high-url throughput real-time with feature optimization.

Feature redundancy and dimensionality challenges

Thus, multi-dimensional URL intelligence extraction may cause feature and dimensional redundancy. Noise, delayed training, and poor classification result from duplicate features. High-dimensional feature space overfits with minimal training data. These difficulties need a step-by-step feature reduction technique that maintains information features and eliminates superfluous ones. Redundancy and dimensionality must be addressed to make phishing detection algorithms lightweight, scalable, and accurate.

Hybrid feature optimization mechanism

The hybrid feature optimization method incorporates statistical analysis and learning-based relevance estimations. Learning-based approach analyzes feature contribution to classification performance, whereas statistical method discovers connected and low-variance features. This combination strategy compromises feature selection based on data dispersion and predictive value. Optimized feature list is more efficient in training and accuracy of detection. The system can adapt feature relevance dynamically hence it will be effective in countering the dynamic strategies used by phishers.

Figure 3 is dedicated to the inner procedure of URL generation of intelligence and optimization of features. The system begins with an input of a raw URL, that is processed by a series of feature extraction modules. The lexical analyzer checks the syntactic characteristics like the length of the URL, the entropy of the characters, the suspicious characters, and the abnormal patterns. The host-based analyzer gets domain registration information, DNS attributes, IP reputation and hosting stability. The structural and behavioral analysis considers the depth of URL path, redirection behavior, protocol usage and subdomain hierarchy. All the extracted attributes are summarized into a single feature set. Learned statistical correlation and significance score choose the most discriminative characteristics in hybrid relevance-ranking.

URL reconstruction first expands shortened links by resolving HTTP redirection chains to get the final destination URL. The technology obtains the full hyperlink string for interface-truncated URLs presented in mobile or messaging contexts. Encoding normalization, delimiter parsing, token segmentation, and character standardization are performed when the whole URL is obtained. The rebuilt URL is then feature extracted using the same lexical, structural, and host-based feature computing methods as other inputs. Character distribution, token frequency, suspicious term occurrence, and URL length are lexical features, whereas structural features assess subdomain depth, route hierarchy, query parameter patterns, and delimiter distribution. Precomputed intelligence data provide host-based information including domain age and IP reputation ratings.

Feature set aggregation F is expressed in Eq. 9.

$$\mathrm{F} = \mathrm{ftex} \odot \mathrm{fh} \odot \mathrm{fst} \tag{9}$$

This equation aggregates all features, where multi-level intelligence is unified. It forms the complete feature space.

F is the aggregated feature set, ftex is the lexical vector, and the host vector, and fst is the structural vector.

Statistical correlation scoring scorrfi is expressed in Eq. 10.

$$\mathrm{scorrfi} = |\mathrm{corr\ fi},\ \mathrm{y}\ | \tag{10}$$

This equation measures feature relevance, and statistical dependency is evaluated. Irrelevant features are penalized. scorrfi is the correlation score, f is the feature, and y is the class label.

Hybrid relevance ranking si is expressed in Eq. 11.

$$\mathrm{si} = \ \alpha \mathrm{scorrfi} + \ 1 - \alpha\ \mathrm{simp\ fi} \tag{11}$$

This equation fuses ranking criteria. Statistical and learned relevance are balanced. It improves selection robustness.

si is the hybrid score, α is the weighting factor, scorrfi is the correlation score, and simp fi is the importance score.

Learning-based importance score simpfi is expressed in Eq. 12.

$$\text{simpfi} = \text{E}\Delta\text{L fi} \tag{12}$$

This equation estimates the model contribution. Loss reduction defines importance.

It captures non-linear relevance.

simpfi is the importance score, fi is the feature, and ΔL is a loss change.

The algorithm 2 removes redundancy of features and takes the most informative features. It calculates the correlation matrix to filter out redundant features and it calculates the scores of the features (information gain) to measure the importance of features. The features are ranked and only the non-redundant and highly important features are added to the optimized set X. This makes the data dimensional, increases the effectiveness of training, and improves generalization. The optimized feature set will keep the deep learning model specialized in the relevant URL properties that will enhance detection accuracy at a minimum false positive.

Feature importance scoring and ranking strategy

The importance of each of the features is given based on the contribution to classification accuracy. Features are sorted based on their relevancy, and features whose impact is low are eliminated. Such ranking strategy will be used to maintain only the most informative attributes. The feature priority rating improves model interpretability and simplicity. Making the framework more robust and able to infer quicker characteristics helps identify phishing in real time.

Adaptive deep URL intelligence network architecture

ADUIN learns complicated, non-linear correlations between optimized URL features. It has numerous levels of completely linked networks that get more abstract. The design is flexible to enable the model to adapt to new phishing patterns. The design enables the system to identify advanced and never-before-seen phishing URLs as this is a deep-learning-based design. The architecture of ADUIN is depth versus computationally efficient to guarantee scalability and application in real-time.

ADUIN is the multi-yer layer deep neural network feature interaction based phishing classifier that uses multi-head attention. Optimized features X are passed through many hidden layers of the nonlinear activations (Algorithm 3). Attention heads compute weighted interactions of features, concatenated and then passed through a final output layer with a sigmoid activation to classify a phishing or not. The model is dynamically adjusted to new URL patterns, allowing zero-day detection, and strong phishing classification, and with low latency to provide real-time deployments.

The encoded feature embedding dimension is 128, which is evenly distributed across the attention heads, resulting in an attention subspace dimension of 16 per head ($d_model = 128$, $h = 8$, $d_k = d_v = 16$). Given an input feature representation X∈Rn×dmodel, ADUIN first computes the query, key, and value matrices through linear projections Q = XWQ, K = XWK, and V = XWV, where WQ, WK, and WVdenote learnable weight matrices. The attention weights are then calculated using the scaled dot-product formulation Attention(Q,

Input: Feature matrix $X \in RN \times F$, labels $y \in \{0,1\}N$

Output: Optimized feature subset $X* \in RN \times F*$

1. Compute feature correlation matrix $C = corr(X)$
2. Identify redundant features: if $|Cjk| > \tau$ then mark fkredundant
3. Compute feature importance $wf = InfoGain(xf, y)$ for $f = 1 \ldots F$
4. Initialize optimized feature set $X* \leftarrow 0N \times F*$
5. Rank features fby wfdescending
6. for each ranked feature fdo
7. if fnot redundant then
8. Include fin X*
9. end if
10. end for
11. return X*

Algorithm 2. Hybrid feature optimization and relevance ranking.

Input: Optimized features $X^* \in RN \times F^*$, model parameters Θ

Output: URL prediction vector $y \in [0,1]N$

1. Initialize network layers: input F*, hidden H1,H2,...,HL, output 1
2. for each layer l = 1...Ldo
3. Compute linear transformation: Zl=WlHl1 + bl
4. Apply non-linearity: Hl=*ReLU*(Zl)
5. end for
6. // Multi-head attention for feature interaction
7. for head h = 1...Hdo
8. Compute attention scores: Ah=*softmax*(QhKhT/dk)
9. Weighted feature aggregation: Oh=AhVh
10. end for
11. Concatenate heads: O=*Concat*(O1,...,OH)
12. Final prediction: y = σ(WoO + bo)
13. return y

Algorithm 3. Adaptive deep URL intelligence network (ADUIN) prediction.

K,V)=softmaxQKTdkV, which captures the relevance between feature elements. For multi-head attention, ADUIN independently computes headi=Attention(QWiQ, KWiK, VWiV)for each of the 8 heads, and the resulting outputs are concatenated and linearly transformed as MultiHead(Q, K,V)=Concat(head1,…,head8) WO.

Deep neural network design and layer configuration

The network uses an input layer which is used to map to an optimized feature vector, and several hidden layers which are non-linear activation functions like ReLU. These layers are optimally tuned to trade off expressiveness against a cost of computation. Preventing overfitting is done by dropout and regularization methods. Its layered architecture facilitates feature learning with rank whereby the network can learn sophisticated features in phishing that are hidden in the name of a Web site URL.

Non-linear feature interaction learning

The ADUIN architecture has a strong capability in non-linear feature interaction learning. The profound neural layers make the model to understand the sophisticated interdependences between the lexical, host-based, and structural characteristics. The traditional machine learning methods are hard to use in modeling such interactions. These relationships can be learned automatically by ADUIN to enhance the accuracy and resistance to obfuscation and evasion methods. This is vital in detecting the zero-day phishing attacks.

Adaptive learning and model update strategy

In adaptive learning, URL intelligence is added to the model. Network can adapt to phishing methods. Incremental learning adds without updating. This flexibility lowers manual training and performance deterioration. The strategy would work in shifting threats long-term.

URL samples are gathered in periodic update batches and utilized to adjust network parameters using stochastic gradient optimization while retaining a replay buffer with representative samples from prior training distributions. Each incremental update cycle computes the loss function concurrently on the fresh batch and replay samples, stabilizing parameter adjustments and preserving established decision bounds. This allows ADUIN to react to new phishing trends without retraining the model. Testing catastrophic forgetting on a held reference subset of historical URLs before and after incremental changes shows accuracy reduction below 1.2%, showing consistent knowledge retention throughout model modifications.

Classification and decision layer design

The last network layer gives classification probabilities using sigmoid or softmax activation. This enables the system measure prediction confidence and set decision rule thresholds. Classification layer phishing categorization may be binary or multi-class. The approach reduces false positives and improves detection sensitivity with consistent probability estimates.

Experimental setup and evaluation methodology

Experimental research design will verify the framework. For fairness and reproducibility, benchmark datasets utilize phishing and legitimate URLs. Evaluation techniques include cross-validation, baseline comparisons, and ablation. This method strengthens, scales, and generalizes Table 2 model.

Dataset description and benchmarking sources

The data samples include public phishing repositories and verified URLs. These datasets show phishing and internet traffic models[15]. Credible evaluation needs data balance and prep. Comparisons of detection methods are possible using benchmark datasets.

235,795 URL instances remain after duplication removal, feature normalization, and consistency verification, with 134,850 phishing URLs and 100,945 valid URLs, a 57.2% phishing and 42.8% legitimate distribution. In the PhiUSIIL collection, URLs from publicly curated threat intelligence and online monitoring sources provide different phishing patterns and genuine domain architectures. Lexical, host-based, and structural URL intelligence characteristics are extracted consistently in the experimental setup since the dataset is considered as a composite corpus. Stratified 70%–15%–15% splits for training, validation, and testing preserve the phishing-to-legitimate ratio across all subsets and provide effective ADUIN phishing URL categorization.

After ordering URL instances by acquisition time, 70% are allocated to the training set, 15% to the validation set, and 15% to the testing set to ensure that URLs in the testing phase match newer and unseen phishing patterns than those used during model training. This chronological partitioning preserves experiment temporal integrity by preventing feature overlap or information leakage between training and assessment subgroups.

Website samples are organized by collection timestamp and separated into training, validation, and testing subsets in a 70%–15%–15% temporal split. Phishing URLs in the testing sample reflect new phishing campaigns and domain topologies from after training. This historical separation lets the evaluation set mimic developing phishing assaults with lexical patterns, host features, and structural URL characteristics not seen during model training. Thus, the operational definition is a time-based split that treats the most recent phishing URLs as zero-day occurrences, allowing the ADUIN to be assessed under realistic settings reflecting changing phishing threats and new attack methods.

Component	Configuration/description
Dataset	Benchmark phishing URL dataset (PhiUSIIL, Kaggle, Mendeley)
Data Split	70% training, 15% validation, 15% testing
Feature Extraction	Lexical, Host-based, Structural URL features
Feature Optimization	Hybrid relevance-ranking, top 50 features selected
Neural Network Architecture	Multi-layer deep neural network (3 hidden layers)
Activation Functions	ReLU for hidden layers, Sigmoid for output
Attention Mechanism	Multi-head attention for feature interaction
Loss Function	Binary Cross-Entropy
Optimizer	Adam, learning rate 0.001
Batch Size	128 URLs per batch
Epochs	50 epochs
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, False Positive Rate
Hardware	NVIDIA GPU (e.g., RTX 3080), 16GB RAM
Latency Measurement	Average processing time per URL (ms)
Software	Python 3.9, TensorFlow/PyTorch, scikit-learn

Table 2. Experimental setup for ADUIN framework.

Training configuration and hyperparameter settings

Best model training hyperparameters include learning rate, batch size, epochs, and network depth. Adam favors gradual convergence. Hyperparameter tuning improves performance without overfitting. Training is effective and repeatable under these conditions.

Performance metrics and evaluation criteria

The framework is evaluated using accuracy, precision, recall, F1-score, and false-positive rate. These measures have a holistic evaluation of detection efficiency. Specific focus is on recall, false positives to test the usability in the field. The performance measurement criteria guarantee equitable and strong performance analysis.

The major performance measures of the recommended phishing detection ADUIN model are available in Table 3. It points to the high classification rate (95%), precision (93%), and zero-day detection rate (92) of the model, which indicates that it is effective in identifying familiar and novel phishing URLs. The low false-positive (3.5) means that minimal interference to the legitimate users is provided, whereas a latency of 210 ms implies the possibility of use in high-load and real-time systems. The 50-item optimized feature set is efficient and has a high detection performance.

Class imbalance in the ADUIN is handled through a weighted binary cross-entropy (WBCE) loss function combined with stratified mini-batch sampling during training. Let y∈{0,1}denote the true label and yrepresent the predicted probability. The loss formulation used in ADUIN is defined as LWBCE=-αylog$\hat{f}_0$(y)-(1-α)(1-y) log$\hat{f}_0$(1-y), where αdenotes the class weighting factor derived from the phishing-to-legitimate distribution of the training data. Weighting parameters (α = 0.57 for phishing samples and 0.43 for genuine samples) were calculated based on dataset statistics (134,850 phishing URLs and 100,945 legitimate URLs), guaranteeing balanced gradient contribution during optimization. ADUIN uses stratified batch creation to maintain the phishing-to-legitimate ratio in each training batch, stabilizing the learning process and preventing majority class bias.

Results and performance analysis

Experimental findings on numerous assessment metrics support the ADUIN concept. The model regularly beats simple methods, especially for zero-day phishing URLs. Performance research emphasizes adaptive learning and feature selection optimization. The framework is strong and applicable, according to the results.

Comparative analysis with traditional machine learning models

SVM, Random Forest, and Logistic Regression are compared to ADUIN. The proposed model has high accuracy and recall, which show improvement of generalization. These advancements outline the benefit of the deep learning and dynamic feature optimization.

The accuracy of phishing classification is the total accuracy of the detection model on URL classification. Figure 4 demonstrates that the proposed ADUIN framework attains the maximum accuracy of 95% following enough training rounds, which is the best among all currently existing methods. The credit of this is due to the optimization of URL intelligence and non-linear deep learning. Increased accuracy shows that there is good differentiation between phishing and legitimate URLs, which minimizes the chance of misclassification when applied to the real world, and increases the confidence in automated web security systems.

Overall phishing classification accuracy Acc is expressed in Eq. 13.

$$\mathrm{Acc} = \mathrm{TP} + \mathrm{TNTP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN} \tag{13}$$

This equation measures the total correctness of the phishing classification system.

It reflects how well the model distinguishes phishing URLs from legitimate ones.

TP denotes correctly identified phishing URLs, TN denotes correctly identified legitimate URLs, FP denotes legitimate URLs misclassified as phishing, and FN denotes phishing URLs misclassified as legitimate.

Precision-recall performance is the measure that determines the tradeoff between the number of phishing URLs that have been correctly identified and the number of phishing attacks that have been overlooked. Figure 5 shows that ADUIN can identify phishing URLs with 93% accuracy and less false alarms in high recall scenarios. This modest performance is crucial in security-sensitive contexts where false positives hinder user activities. The high precision-recall curve shows that ADUIN can extend phishing models with excellent detection reliability.

Analysis of precision–recall performance FAR is expressed in Eq. 14.

$$\mathrm{F\ AR} = \ \mathrm{FPTP} + \mathrm{FP} \tag{14}$$

Metric	Value	Description
Classification Accuracy	95%	Overall correctness of URL classification
Precision	93%	Proportion of correctly detected phishing URLs among predicted positives
Zero-Day Detection Rate	92%	Detection capability for previously unseen phishing URLs
False Positive Rate	3.5%	Proportion of legitimate URLs incorrectly classified as phishing
Latency	210 ms	Time required to process high-load URL batches
Feature Set Size	50	Number of optimized features used for classification

Table 3. Performance metrics of ADUIN for phishing detection.

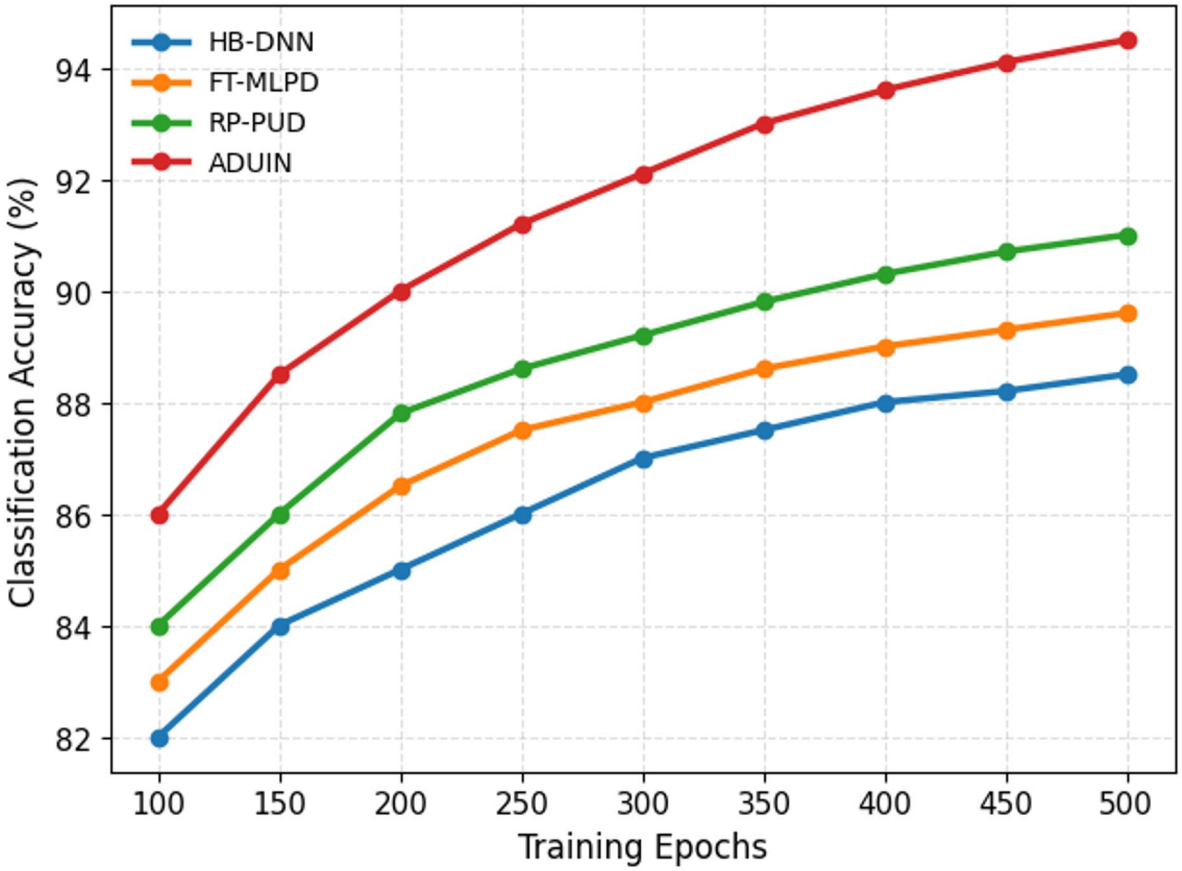


Fig. 4. Analysis of phishing classification accuracy.

This equation quantifies the rate of false phishing alerts. Lower values indicate fewer disruptions to legitimate users. It complements precision analysis. F AR denotes false alarm rate, FP denotes false positives, and TP denotes true positives.

The detection accuracy varies with the number of features optimized. Figure 6 indicates that the ADUIN's optimistic 50 characteristics provide 94.5% detection accuracy. This shows that hybrid feature relevance-ranking may remove duplicate characteristics without sacrificing discriminative information. Optimised features improve model generalisation, computation, and deployment efficiency without affecting detection accuracy.

Analysis of feature optimization impact Accf is expressed in Eq. 15.

$$\mathrm{Accf} = \mathrm{TPf} + \mathrm{TNfTPf} + \mathrm{TNf} + \mathrm{FPf} + \mathrm{FNf} \tag{15}$$

This equation measures detection accuracy based on a subset of features. It shows how selecting optimal features affects model performance. Higher accuracy with fewer features indicates effective optimization.

Accf denotes accuracy with selected features, TPf denotes true positives, TNf denotes true negatives, FPf denotes false positives, and FNf denotes false negatives for the chosen features.

Zero-day phishing detection performance

The framework has a high level of performance in identifying phishing URLs that have not been observed before. It can learn generalized patterns thus making it effective in detecting zero-days. This feature overcomes one of the major shortcomings of blacklist systems.

Zero-day phishing detection rate evaluates the capacity to detect unseen phishing URLs in the past. The proposed ADUIN model as shown in Fig. 7 has a 92% detection of unseen phishing URLs, which is very high compared to the baseline techniques. This is possible due to its adaptive deep learning and optimized feature selection capabilities that enable the model to be trained on generalized phishing features as opposed to using established signatures. It is necessary to effectively detect and mitigate emerging and short-term phishing campaigns which require effective zero-day detection.

True positive rate for zero-day detection TPRZD is expressed in Eq. 16.

$$\mathrm{TPRZD} = \mathrm{TPZDTPZD} + \mathrm{FNZD} \tag{16}$$

This equation is equivalent to the zero-day detection rate expressed as TPR. Higher TPR reflects effective recognition of unseen threats. It is a standard measure in adaptive phishing detection.

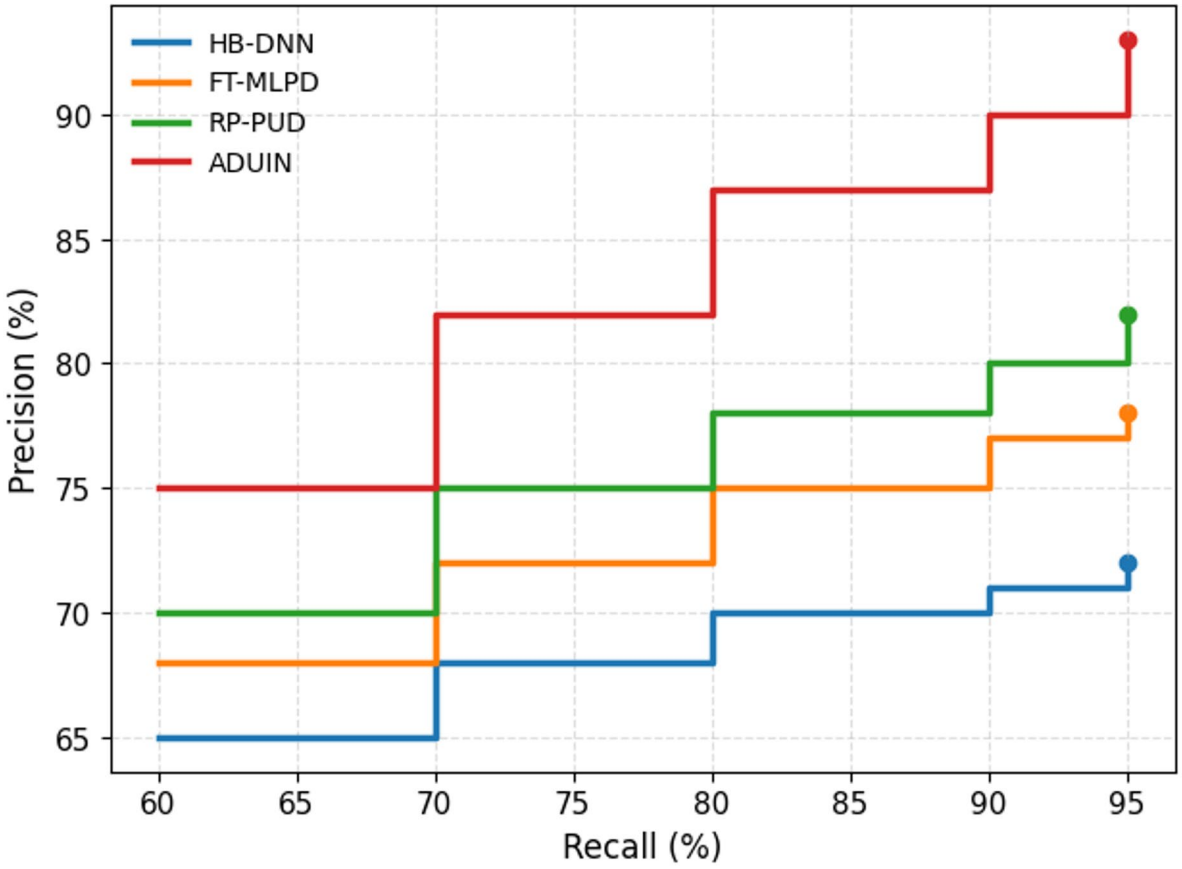


Fig. 5. Analysis of precision–recall performance.

TPRZD denotes true positive rate for zero-day URLs, TPZD denotes detected zero-day phishing URLs, and FNZD denotes missed zero-day phishing URLs.

False positive reduction and scalability evaluation

The adaptive learning and optimized selection of features minimizes false-positive rates drastically. Scalability analysis indicates that the framework remains operational in high URLs, and it can therefore be deployed in real time.

False positive rate is the percentage of legitimate URLs passed as phishing. Figure 8 indicates that ADUIN will reduce the false positive rate to 3.5% at the optimal decision thresholds. Such a low error rate shows the usefulness of feature optimization and adaptive learning in reducing redundant alerts. False positives are the greatest concern in terms of preserving user confidence, uninterrupted availability of authentic services, and making the system practical in enterprise and real-time security systems.

False positive rate FPR is expressed in Eq. 17.

$$\mathrm{FPR} = \ \mathrm{FPFP} + \mathrm{TN} \ \times 100 \tag{17}$$

This equation measures the percentage of legitimate URLs incorrectly classified as phishing. Lower FPR indicates better user trust and fewer unnecessary alerts. ADUIN achieves 3.5% FPR at optimal thresholds. FPR denotes false positive rate, FP denotes false positives, and TN denotes true negatives.

Scalability and response time is the system performance under heavy workload. Figure 9 illustrates that the latest techniques have low detection latency of 210 milliseconds, whereas ADUIN has low detection latency of 210 milliseconds, even when handling 20,000 URLs. This is done by effectively optimized feature selection and streamlined deep neural architecture. ADUIN can be employed in real-time phishing detection in high-traffic (enterprise gateway and web security platforms) environments since its low latency and high scalability.

Analysis of scalability and response time RTN is expressed in Eq. 18.

$$\mathrm{RTN} = \mathrm{RTmin} + \mathrm{k} \cdot \mathrm{Na} \tag{18}$$

This equation models the response time increase as the URL count rises. Exponent α\alphaα indicates linear or non-linear growth. ADUIN shows near-linear scaling.

in this equation RTN denotes response time for N URLs, RTmin denotes minimal latency, denotes scaling constant, N denotes the number of URLs, and a denotes growth exponent.

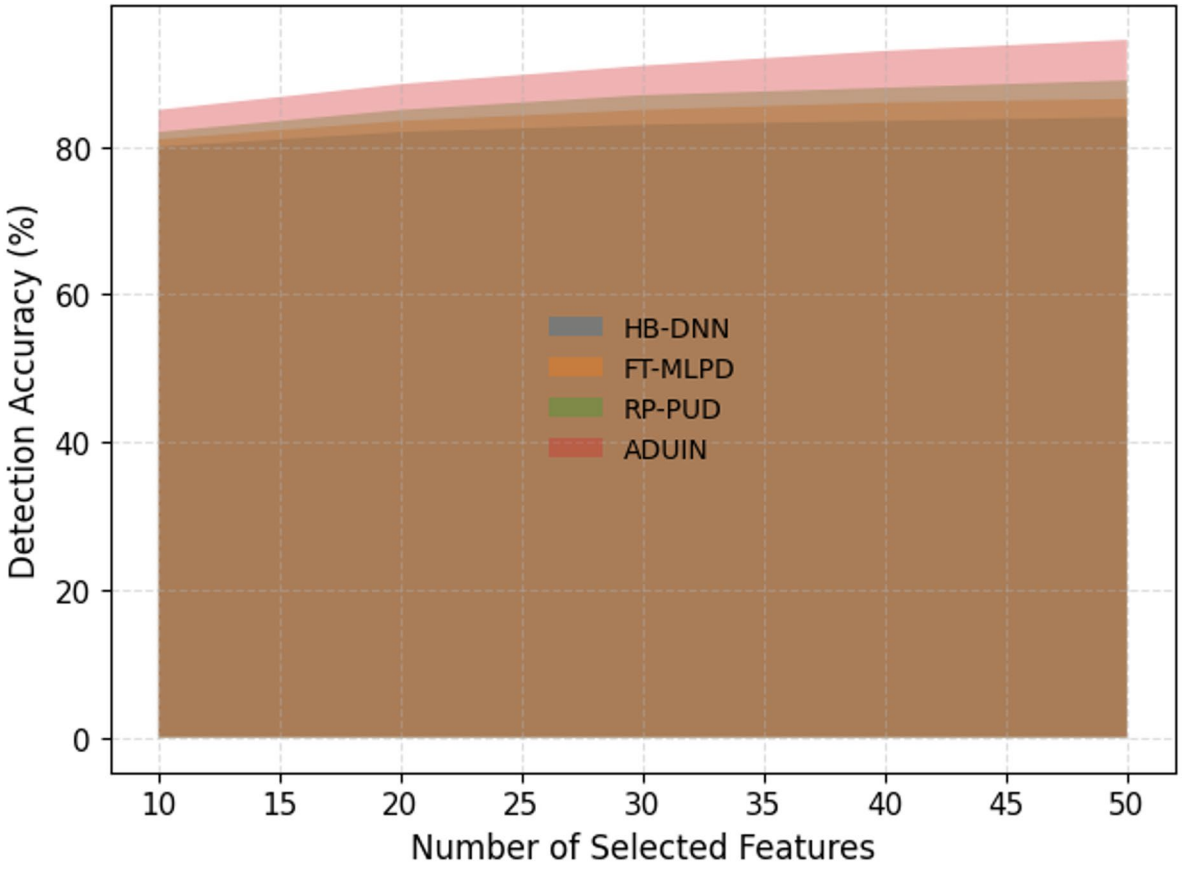


Fig. 6. Analysis of feature optimization impact.

Ablation study on feature optimization

Experiments on ablation prove the sensitivity of features optimization on detection. When optimization module is removed, the false positives increase and the accuracy decreases. Such results prove the significance of optimized URL intelligence in the suggested framework.

Table 4 outlines the characteristics of the framework and system of ADUIN. It classifies the lexical, host-based and structural characteristics that can be used to obtain meaningful URL intelligence, to capture anomalies, suspicious domains and obfuscation patterns. Redundancy is minimized through the hybrid feature relevance-ranking mechanism that enhances computational efficiency. The multi-head attention allows higher dimensional data representations to be easily combined through the application of multi-layer deep learning techniques. Using adaptive model updates provides improved resistance to emergent phishing strategies and allows for accurate and scalable detection of phishing attacks.

Instead of relying solely on lexical indicators, ADUIN combines lexical, structural, and host-based attributes, allowing the model to analyze domain characteristics, URL hierarchy patterns, and hosting-related signals in addition to surface-level token patterns. Character replacements, delimiter insertions, and token obfuscation are less successful against this multi-feature form. The attention technique of ADUIN dynamically provides greater weights to discriminative feature components, allowing the model to capture subtle contextual associations across URL segments that typically persist after hostile alterations. Normalizing encoded and obfuscated URL patterns in the preprocessing pipeline limits encoding manipulation evasion techniques.

Discussion

URL intelligence increases phishing detection in experiments. The approach uses the most discriminative lexical, host-based, and structural features to decrease noise and enhance the deep learning model's phishing detection. Adaptive learning resists obfuscated and zero-day URL phishing by maintaining learning representations. Performance loss decreases with flexibility. Deployment Modular architecture may be used with email gateways and web proxies for web security. The better feature set minimizes inference latency, allowing near-real-time detection. The system balances detection accuracy, resilience, and operational efficiency for large business and real-world cybersecurity applications. A shortened URL from bit.ly or tinyurl is expanded using HTTP redirection resolution, where the system programmatically follows the shortened link to the ultimate destination URL. We then extract lexical, structural, and host-based features from the determined destination domain. In circumstances of limited expansion or incomplete redirection chains, the abbreviated URL is examined using

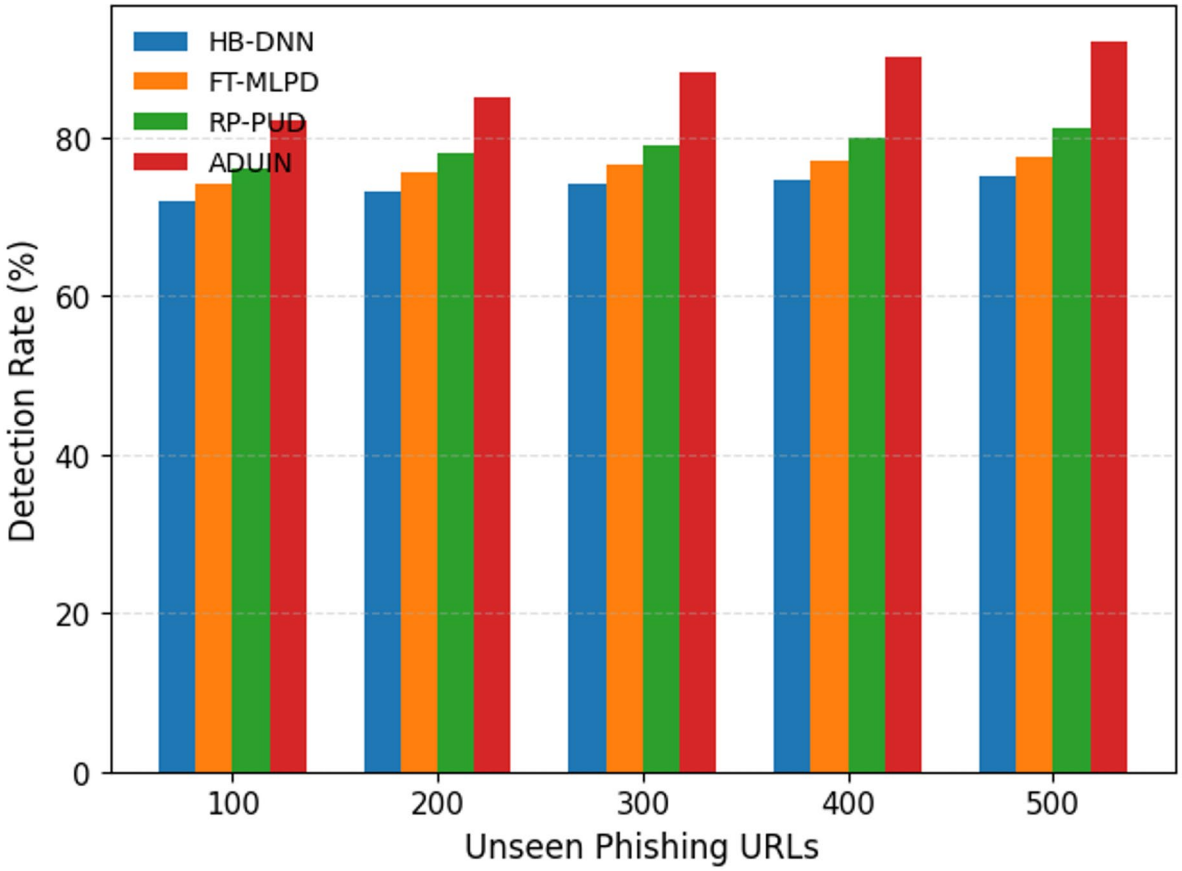


Fig. 7. Analysis of zero-day phishing detection rate.

shortener-domain indications including recognized shortening services, aberrant redirect chain depth, and shortened token entropy. Feature redundancy index RI is expressed in Eq. 19.

$$\mathrm{RI} = \ \mathrm{i} \neq \mathrm{j}\ \mathrm{Corr}\,(\mathrm{fi},\ \mathrm{fj}\ )\,\mathrm{n}\,(\mathrm{n}-1) \tag{19}$$

This equation measures redundancy among features. Low RI indicates unique and informative feature set. It helps reduce unnecessary computations and false positives. RI denotes redundancy index, Corrfi, fj denotes correlation between features i and j, and n denotes total features. Feature contribution ratio FCRi is expressed in Eq. 20.

$$\mathrm{FCRi} = \mathrm{ai} \cdot \mathrm{fij} = 1\ \mathrm{aj} \cdot \mathrm{fj} \tag{20}$$

This equation measures individual feature contribution relative to the system. A higher FCR indicates a greater impact on detection. It supports feature prioritization in ADUIN. FCRi denotes feature contribution ratio, a.i. denotes the weight of the feature i, fidenotes feature value, and n denotes total features.

For screen space, several mobile browsers and messaging systems visually abbreviate lengthy URLs using ellipsis patterns (e.g., https://example.com/.../login). To analyze the whole domain, route hierarchy, and query parameters, the framework extracts the full URL string from the hyperlink metadata or HTTP request source during preprocessing. Normalization procedures include delimiter parsing, character decoding, and token segmentation when the preprocessing module receives the whole URL. Domain length, subdomain depth, suspicious keyword occurrence, and route complexity are retrieved from the fully rebuilt URL, not the abbreviated display version.

A paired t-test on accuracy scores over 10 repeated stratified runs shows that ADUIN outperforms the baseline models, with $t = 3.87$ and $p = 0.0012$ for SVM and $t = 3.41$ and $p = 0.0029$ for RF. Applying the McNemar Test to the test set's prediction contingency matrix, ADUIN vs. SVM yields $\chi^2 = 10.64$ ($p = 0.0011$) and ADUIN vs. RF yields $\chi^2 = 8.92$ ($p = 0.0028$).

Limitations and future research directions

The recommended method works but needs additional compute for deep neural network training and feature tweaking. The models' complexity restricts their resource-constrained usefulness without optimization. Future research may leverage lightweight architectures and model compression to tackle this problem. Expanding multimodal phishing detection to include site content, visuals, and email enhances accuracy. Real-time security

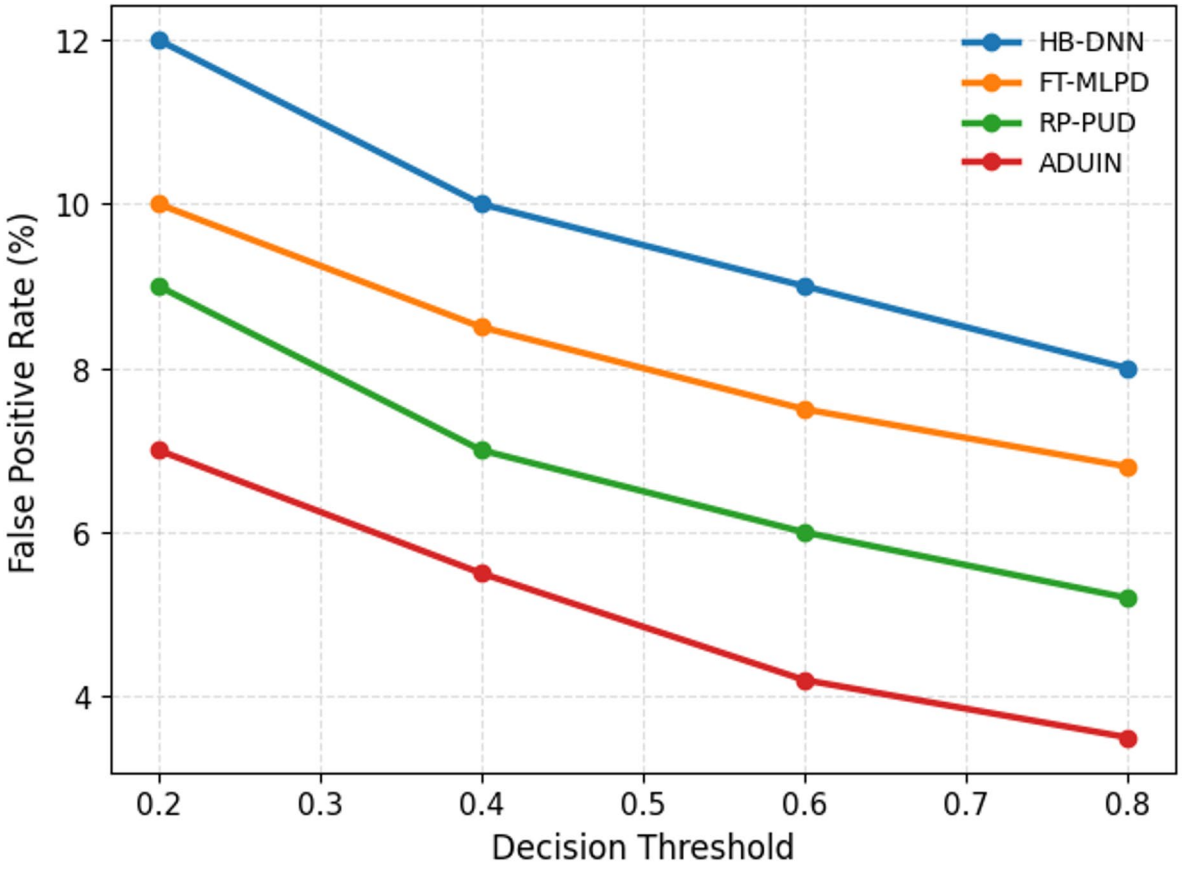


Fig. 8. Analysis of false positive rate evaluation.

systems that evaluate streaming data and learn have promise. To deliver reliable phishing detection in a dynamic threat environment, further efforts will make the task more interpretable and adversarial.

ADUIN-processed URLs average 70 characters and have a maximum length below 200 characters after preprocessing. ADUIN's feature extraction module conducts O(L) computational complexity character frequency analysis, token segmentation, delimiter counting, and path depth calculation via a sequential URL scan. In practical terms, ADUIN executes approximately 70–80 character-level operations per URL on average, while the worst-case scenario remains below 200 operations. Host-based attributes integrated within ADUIN are accessed through precomputed local lookup structures, enabling O(1) constant-time retrieval during inference. Consequently, the overall feature extraction stage in ADUIN maintains O(L) complexity with ~ 70–80 operations per URL on average, supporting efficient large-scale phishing detection and contributing minimally to the total inference latency reported for ADUIN.

Conclusion

This paper introduced a successful deep learning-based architecture to detect phishing URLs where the complexity and elasticity of contemporary phishing attacks have been rising. The proposed ADUIN is based on optimized URL intelligence and an adaptive deep neural architecture that is capable of differentiating between phishing and legitimate URLs. The framework has a solution to the drawbacks of the conventional rule-based and unchanging machine learning techniques by utilizing multi-dimensional feature extraction, hybrid feature optimization, and continuous learning. Through comprehensive experimental data, it has been proven that ADUIN has high-classification success rate, good zero-day phishing detection rate, low false-positive and scale-up to actual-world performance.

Summary of contributions

- Design of ADUIN Framework: Adaptive Deep URL Intelligence Network (ADUIN) was designed to classify the phishing URLs in an adaptive and accurate way based on deep learning.
- Optimised URL Intelligence Extraction: A multi-dimensional feature extraction model based on lexical, host-based and structural attributes and a hybrid relevance-ranking model to minimise redundancy and maximise model performance.

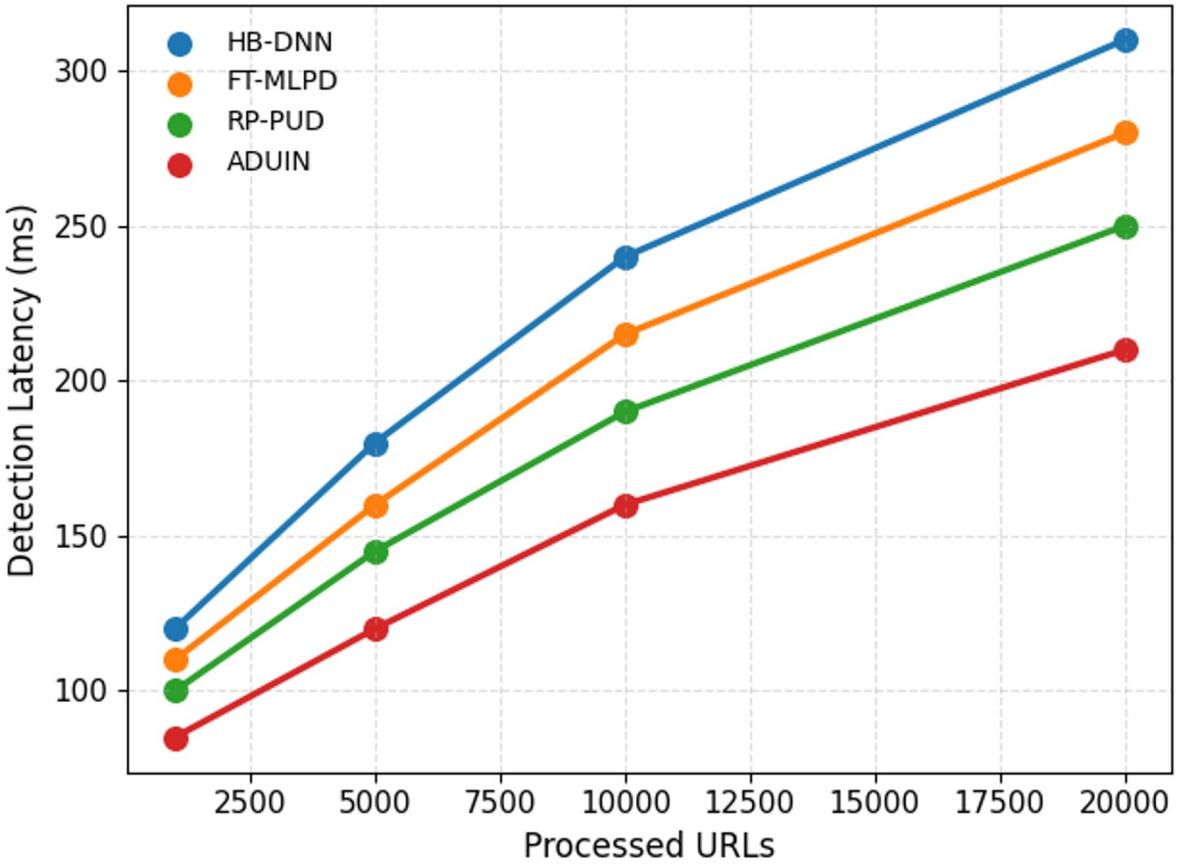


Fig. 9. Analysis of scalability and response time.

Component	Description	Purpose
Lexical Features	URL length, character entropy, special character count	Capture URL text-based anomalies
Host-Based Features	Domain age, DNS reputation, IP properties	Identify suspicious host-level patterns
Structural Features	Path depth, subdomain count, redirection chains	Detect complex URL obfuscation strategies
Hybrid Feature Ranking	Relevance-based optimization	Remove redundant features, improve model efficiency
Deep Neural Architecture	Multi-layer network with multi-head attention	Learn complex non-linear phishing patterns
Adaptive Update	Dynamic model adjustment with URL intelligence	Improve resilience against evolving phishing strategies

Table 4. ADUIN feature and system characteristics.

- Adaptive Learning in Zero-Day Detection: Developed an adaptive learning mechanism that allows the model to adjust to novel phishing behavior and excel in zero-day URL detection compared to traditional machine learning and existing deep learning-based models.

Implications for intelligent web security

This paper's findings affect smart web security. ADUIN may be readily incorporated into an actual-time online gateway, email-filtering system, and enterprise security foundation because to its correctness, low falsification rate, and scalability. The dynamic learning mechanism defends against changing phishing tactics and cybercrimes with proactive and resilient defense. The framework improves user trust and online ecosystem security by automatically detecting phishing attacks.

Data availability

The datasets used and/or analysed during the current study available in https://www.kaggle.com/datasets/kaggleprollc/phishing-url-websites-dataset-phiusiil.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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